Siril & Seti Astro Suite Pro Processing Companion

Dedication

Dedicated to my beloved wife Donna.

Preface

When I first pointed a telescope toward the night sky in the mid-1960s, I never imagined that astrophotography would become a lifelong pursuit. My journey began with a Unitron 3-inch equatorial refractor and an Unitron Model 220 Astrocamera sheet film camera. In those days there were no computers, no instant image review, and no artificial intelligence. Every exposure demanded patience, and every developed frame taught a lesson, often hard.

During the 1970s and 1980s I moved to 35 mm astrophotography using Kodak Technical Pan (Tech Pan) film. Long manually guided exposures under dark skies fostered an appreciation for careful technique and respect for every photon captured. In the 1990s I embraced the digital era by building a Cookbook CCD camera (TY Bob Higgins), joining countless amateur astronomers who ushered in a revolution in electronic imaging.

Today, in my seventies, I remain amazed by the extraordinary advances in our hobby. Modern CMOS cameras, precision mounts, plate solving, robotic observatories, and AI-assisted processing have transformed what is possible for amateur astronomers. Yet the sense of wonder has never changed.

I wanted to create a roadmap for myself to understand the data I use to realistically reveals the beauty of God's universe that traveled across space to reach my telescope.

Acknowledgements

To the amateur astronomy community, whose willingness to share ideas and offer help has advanced the hobby for generations. To the developers of Siril, Seti Astro Suite Pro, RC-Astro, SyQon, VeraLux, GraXpert, and countless other tools that continue to expand the possibilities of amateur imaging.

Introduction

This companion is a gathering from the work of many other people and my own personal experiences. I tried to organize this around already established principles. Every workflow emphasizes understanding the data, choosing the appropriate processing path, validating each major decision, and producing images that balance scientific integrity with visual impact. A great deal of time was spent creating a workflow based on my own experiences over the years, and most importantly I gleaned many processing gems form the sources reference at the end of this companion. I am not recreating anything, rather I organized my copious notes (gathered over years) with the best practices of others into a text file draft then AI compiled it into an easy-to-follow format.

Chapter 1 – The Philosophy of Astrophotography Processing

Why We Process

Image processing is not the act of manufacturing beauty or artificially creating data that did not exist—it is the disciplined process of revealing the information already present in astronomical data. Every adjustment should help the viewer see more clearly what the telescope recorded.

The Workflow Companion Philosophy

- Maximum Aesthetic Impact while preserving realism.
- Separate sharpening and denoising into independent stages.
- Validate every major processing step.
- Use the best tool for the task rather than committing to a single software ecosystem.
- Preserve the archival master before presentation exports.

Recommended Tools and Alternatives

Processing Stage	Recommended	Alternatives	When to Choose
Background Extraction	AutoBGE	GraXpert AI, VeraLux Nox, Native	General use vs. difficult gradients
Sharpening	RC-Astro BlurX	SyQon Parallax, Native Richardson-Lucy	AI recovery vs. classical control
Denoising	RC-Astro NoiseX	SyQon Prism, VeraLux Silentium, Native	Balance noise and faint signal
Star Management	RC-Astro StarX	StarNet, Native	Only when stars distract from the target
Color	SPCC	PCC, Native White Balance	Prefer photometric calibration when possible
Finishing	VeraLux Revela & Vectra	CLAHE, Native Curves	Subtle local contrast and color refinement

Validation Checklist

- Background free of gradients.
- Histogram not clipped.
- Natural star colors.
- No sharpening halos.
- Faint structures preserved.

- Image remains believable.
- Validate the Image Before Proceeding
- Before moving to the next processing step, pause and carefully inspect the image. Validation is your opportunity to catch problems before they become difficult—or impossible—to correct.
- Checklist:
- Histogram: Verify that neither the black point nor the highlights are clipped.
- Background: Ensure the background remains smooth and neutral without introducing blotchy gradients or color casts.
- Stars: Examine stars at 100% magnification for halos, ringing, distorted profiles, or loss of natural color.
- Fine Detail: Confirm that faint dust lanes, IFN, and subtle nebular structures have been enhanced—not erased—by sharpening or noise reduction.
- Noise: Look for over-smoothing that produces a plastic appearance or aggressive sharpening that creates artifacts.
- Color: Check that star colors remain natural and that the object's color balance is believable rather than oversaturated.
- Multiple Zoom Levels: Evaluate the image both at Fit to Screen (overall composition and balance) and at 100% (critical detail).
- Comparison: Whenever possible, compare the current image with the previous saved version to ensure each processing step has improved the image without sacrificing real astronomical signal.
- Only after the image passes these checks should you continue with the next stage of processing.

• **Final Processing:** Before saving the 16-bit TIFF master, inspect the full image at Fit view, examine fine details at 100% zoom, and check the histogram for clipped shadows or highlights.

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Notes

Technology changes, but the fundamentals do not. Whether exposing sheet film in 1965 or processing CMOS data with AI today, success still comes from understanding the sky, collecting good data, and exercising restraint during processing.

Check the histogram—preferably with individual RGB channels visible—to verify that important shadow and highlight data is not clipped against the left or right edges.

Chapter 2 – Understanding Your Data

Introduction

Exceptional processing begins long before the first stretch. The quality of the final image is fundamentally limited by the quality of the captured data. Learning to evaluate astronomical data objectively is one of the most valuable skills an astrophotographer can develop.

The Nature of Astronomical Signal

Every pixel contains a mixture of desired signal and unwanted noise. Processing cannot create information that was never captured; it can only reveal or preserve existing signal.

Calibration Frames

Bias, dark, flat, and dark-flat frames each remove a different class of systematic error. Accurate calibration improves every later processing stage.

Dynamic Range

Bright stars, faint dust, and background sky often coexist in the same image. Successful processing preserves all three whenever possible.

Evaluating Data Quality

Inspect focus, star shapes, gradients, tracking, satellite trails, and histogram position before beginning any enhancement.

Recommended Tools and Alternatives

Task	Recommended	Alternatives	When to Choose
Gradient Removal	AutoBGE	GraXpert AI, VeraLux Nox, Native	AutoBGE for most images; GraXpert for severe gradients.
Inspection	Native Statistics	Histogram tools	Verify data before processing.
Color Calibration	SPCC	PCC, Native White Balance	Use photometric calibration whenever possible.
Noise Evaluation	Native analysis	RC-Astro, SyQon previews	Assess before denoising.

Validation Checkpoint

- No clipped histogram.
- Gradients identified.
- Calibration artifacts removed.
- Stars round and well focused.

- Background evaluated before stretching.

Notes

With film, we often waited days to discover whether an exposure succeeded. Modern software gives immediate feedback, but it also tempts us to process weak data too aggressively. Good acquisition still matters more than clever processing.

Compare two integrated masters: one with excellent calibration and one with intentionally omitted flats. Identify the differences before applying any enhancement.

Chapter 3 – Building Your Processing Strategy

Introduction

There is no single perfect workflow. The best workflow is the one that matches the target, the data quality, and the imaging goals. This chapter introduces a structured decision-making process.

Choose a Workflow Profile

Select Fast Preview, Standard, Premium, Maximum Aesthetic, or Scientific based on the purpose of the image.

Target-Based Decisions

Galaxies emphasize dust lanes, emission nebulae emphasize faint filaments, IFN demands extreme restraint, and globular clusters prioritize stellar fidelity.

Separate Processing Stages

Treat stretching, sharpening, denoising, star management, and finishing as independent decisions rather than a fixed recipe.

Validate Frequently

After each major stage, inspect the image at multiple zoom levels and compare with the previous version.

Workflow Decision Matrix

Stage	Preferred	Alternatives	Decision
Sharpen	RC-Astro BlurX	SyQon Parallax, Native Deconvolution	Choose AI for convenience, native for precise control.
Denoise	RC-Astro NoiseX	SyQon Prism, VeraLux Silentium, Native	Use the lightest effective touch.
Stars	RC-Astro StarX	StarNet, Native	Only if it improves the subject.
Finish	Revela + Vectra	CLAHE, Native Curves	Refine rather than transform.

Common Mistakes

- Using the same workflow for every target.
- Sharpening before evaluating noise.
- Applying excessive saturation.
- Ignoring validation checkpoints.

Notes

Over six decades I've learned that the best processors are not always those with the most software; they are the ones who know why they are making each adjustment (that is my goal).

Key Takeaways

- Build the workflow around the data.
- Choose tools objectively.
- Validate after every major stage.
- Preserve the archival master.

Chapter 4 – Linear Processing: Building the Best Foundation

Introduction

Linear processing is the most critical stage of the workflow because every decision made here affects all later enhancement. The objective is to preserve every photon of real signal while removing gradients, calibration artifacts, and color imbalance.

What Is Linear Data?

Linear data preserves the original proportional relationship between recorded photons and pixel values. This is the stage where scientific integrity is greatest and where careful processing yields the largest downstream benefits.

Background Extraction Philosophy

Background extraction should remove unwanted sky gradients without removing genuine astronomical signal. Conservative settings are almost always preferable to aggressive correction.

Color Calibration

Photometric calibration (SPCC when available) is the preferred method because it uses reference star colors. PCC and manual white balance remain useful alternatives in appropriate situations.

Linear Validation

Before stretching, verify that gradients are controlled, the histogram is intact, no clipping has occurred, and faint structures remain visible.

Tool Comparison

Stage	Preferred	Alternatives	Best Use
Background	AutoBGE	GraXpert AI, VeraLux Nox, Native	General vs difficult gradients
Color	SPCC	PCC, White Balance	Accurate stellar color
Inspection	Histogram/Statistics	Native analysis	Verify before stretch

Validation Checklist

- Background neutral
- No clipping
- Stars remain natural
- Faint dust retained
- Ready for nonlinear processing

Notes

Film taught patience. Linear processing teaches discipline. The better the foundation, the less work is required later.

Compare AutoBGE and GraXpert AI on the same dataset. Record differences in faint nebulosity and background neutrality, then choose best fit.

Chapter 5 – Nonlinear Processing: Revealing the Hidden Image

Introduction

Stretching transforms a linear image into one that the human eye can appreciate. The challenge is revealing faint detail without sacrificing highlight control or introducing artifacts.

Stretch Philosophy

Multiple gentle stretches generally preserve more detail than one aggressive transformation.

Choosing a Stretch

Generalized Hyperbolic Stretch (GHS) provides exceptional control over highlights and shadows. Statistical Stretch often excels with faint nebulae. Asinh and other methods remain useful for specialized applications.

Dynamic Range Management

Protect bright galaxy cores, stellar highlights, and planetary nebula interiors while gradually revealing faint structures.

Validation

After stretching, inspect the histogram, compare against the linear master, and evaluate at Fit, 100%, and 200% zoom.

Stretch Decision Matrix

Method	Strengths	Limitations	Recommended Targets
GHS	Excellent control	Learning curve	Galaxies, mixed targets
Statistical Stretch	Preserves faint signal	Less highlight control	Emission & IFN
Asinh	Excellent star color	Limited flexibility	Wide-field OSC
Hyperbolic/Native	General purpose	Varies by implementation	Routine processing

Common Mistakes

- Stretching too aggressively.
- Clipping the black point.
- Trying to offset weak acquisition data through processing.
- Beginning sharpening before evaluating the stretch.

Notes

Patience at this stage pays dividends throughout the remainder of the workflow.

Key Takeaways

- Preserve the histogram.
- Stretch gradually.
- Choose the method to match the target.
- Validate stretch quality, histogram, stars, and faint detail before moving to enhancement.

Chapter 6 – Detail Enhancement: Recovering Resolution Naturally

Introduction

Sharpening is the art of revealing legitimate detail rather than inventing it. The best sharpening algorithms improve local contrast while preserving realistic star profiles and avoiding halos or ringing.

Historical Perspective

Film astrophotography relied on optics and fine-grain emulsions for detail. CCD imaging introduced deconvolution, while modern AI tools can recover subtle structure with remarkable effectiveness when used conservatively.

Recommended Workflow

Perform sharpening only after a successful nonlinear stretch. Evaluate the image before sharpening, apply the lightest effective enhancement, then re-evaluate before continuing.

Decision Philosophy

Use AI sharpening when convenience and robust results are desired. Choose classical deconvolution when maximum control over point spread function is required.

Sharpening Tool Comparison

Tool	Strengths	Limitations	Best For	Notes
RC-Astro BlurX	Excellent AI recovery	Commercial	General use	Recommended default
SyQon Parallax	Fine filament preservation	Learning curve	Nebulae	Excellent alternative
Richardson-Lucy	Scientific control	Requires tuning	High-SNR data	Native option
Native Deconvolution	Integrated workflow	Less forgiving	Routine work	No extra software

Validation Checklist

- No halos around stars
- Fine detail improved
- Noise not amplified excessively
- Natural appearance maintained

Notes

The temptation is always to sharpen just a little more.

Chapter 7 – Noise Reduction: Preserving Signal While Reducing Noise

Introduction

Noise reduction should never become detail reduction. The objective is to suppress random noise while protecting faint galaxies, dust lanes, and nebular filaments.

Understanding Noise

Noise originates from photon statistics, electronics, thermal effects, and the atmosphere. Longer integration always outperforms aggressive denoising.

Workflow Strategy

Evaluate the background first. Apply denoising conservatively and independently from sharpening. Consider masks for protecting high-detail regions.

Scientific Considerations

Every denoising algorithm removes some information. The goal is to maximize the signal-to-noise ratio while minimizing the loss of genuine structure.

Noise Reduction Tool Comparison

Tool	Strengths	Limitations	Best For	Notes
RC-Astro NoiseX	Outstanding AI preservation	Commercial	General use	Recommended default
SyQon Prism	Excellent faint signal retention	Learning curve	Nebulae & IFN	Strong alternative
VeraLux Silentium	Smooth backgrounds	Less common	Broadband	Useful finishing tool
Native Noise Reduction	Integrated workflow	More manual tuning	Routine work	Good baseline

Understanding the Noise Reduction Decision Matrix

There is no single noise reduction method that is ideal for every image. The optimal choice depends on the target, the signal-to-noise ratio (SNR), the amount of faint detail present, and your processing goals. The following recommendations provide a starting point based on the characteristics of the data.

Situation	Preferred	Alternative
High-SNR Galaxy	Light NoiseXTerminator (RC-Astro)	Native masked denoise
Emission Nebula	SyQon Prism or NoiseXTerminator (RC-Astro)	VeraLux Silentium
IFN / Dark Nebula	Minimal NoiseXTerminator (RC-Astro) or SyQon Prism	Mask-protected native denoise
Broadband OSC	NoiseXTerminator (RC-Astro)	Native denoise

High-SNR Galaxy

High signal-to-noise galaxy images generally require only modest noise reduction. The goal is to preserve delicate spiral arms, dust lanes, tidal streams, and fine galactic structure while removing only the remaining background noise. Excessive denoising can soften the very features that make galaxies visually interesting.

Recommendation: Apply NoiseXTerminator (RC-Astro) conservatively using light settings.

Alternative: A carefully masked native denoise tool works well when only small areas of the image require treatment.

Emission Nebula

Emission nebulae often contain intricate filaments and very low-contrast hydrogen and oxygen structures. These features benefit from AI-based denoising that preserves small-scale detail while reducing background noise.

Recommendation: Use SyQon Prism or NoiseXTerminator, depending on the character of your data and personal preference.

Alternative: VeraLux Silentium offers another effective approach, particularly when a softer, more natural appearance is desired.

Integrated Flux Nebula (IFN) and Dark Nebulae

IFN represents some of the faintest structures captured by amateur astronomers. Heavy noise reduction can easily erase these subtle dust clouds or reduce them to an artificial, plastic appearance.

Recommendation: Apply minimal NoiseXTerminator or SyQon Prism using conservative settings. Inspect the image frequently at both Fit-to-Screen and 100% magnification.

Alternative: A carefully protected native denoise process using luminance or range masks can preserve extremely faint dust while limiting smoothing to the background.

Broadband OSC Images

Broadband one-shot color images often contain moderate chrominance and luminance noise, particularly when acquired under light-polluted skies. AI-based denoising provides an excellent balance between noise reduction and detail preservation.

Recommendation: NoiseXTerminator is an excellent default choice for most broadband OSC datasets.

Alternative: Native denoise tools remain effective for users who prefer a traditional processing workflow or who require maximum manual control.

Common Mistakes

- Applying denoise before evaluating the stretch.
- Using one global setting everywhere.
- Removing faint dust with heavy smoothing.
- Confusing residual gradients with noise.

Notes

Noise reduction should be viewed as the last step in improving image quality—not the first. If your image requires aggressive denoising, first examine your acquisition process. Better calibration frames, longer total integration time, darker skies, and improved guiding almost always produce better results than increasing the strength of any noise reduction tool. My rule is simple: apply the minimum amount of denoising necessary to improve the image while preserving every bit of real astronomical signal.

Key Takeaways

- Longer integration beats heavier denoising.
- Sharpening and denoising are separate decisions.
- Protect faint structures.
- Always compare with the pre-denoise version.

Chapter 8 – Star Management: Supporting the Subject, Not Dominating It

Introduction

Stars provide context, scale, and beauty, but they can also overwhelm faint nebulae and galaxies. The purpose of star management is to improve visual balance while preserving a natural appearance.

The Philosophy of Star Management

Stars are part of the astronomical scene, not defects to be removed. Temporary star removal is a processing technique—not an artistic requirement.

When to Use Starless Processing

Dense Milky Way fields, emission nebulae, and complex dust structures often benefit from temporary star removal. Globular clusters and open clusters generally do not.

Recombination Strategy

Process stars and the deep-sky object independently, then recombine with restraint to preserve realistic star colors and sizes.

Star Management Tool Comparison

Tool	Strengths	Limitations	Best Use	Recommendation
RC-Astro	Excellent star separation	Commercial	General purpose	Primary recommendation
StarNet	Widely used	Occasional artifacts	General purpose	Strong alternative
Native Seti Astro	Integrated workflow	Feature dependent	Routine work	Convenient
Native Siril	Basic star tools	Less automated	Simple workflows	Good baseline

Validation Checklist

- Star colors remain natural.
- No dark halos after recombination.
- Star sizes remain believable.
- Target receives emphasis without looking artificial.

Notes

For decades our only option was to process everything together. Modern star-separation tools are remarkable, but the best compliment is when no one notices they were used.

Chapter 9 – Color, Contrast, and Final Finishing

Introduction

Final finishing is where an image gains polish, not where it is rescued. The strongest images require only subtle refinement after a solid workflow.

Color Philosophy

Accurate color calibration should be established early. Final color grading should enhance, not replace, physical realism.

Local Contrast

Use local contrast to reveal texture in galaxies and nebulae while avoiding exaggerated edges and halos.

Preparing for Output

Maintain a 16-bit archival master. Create separate exports for web, print, and publication rather than relying on one file for every purpose.

Finishing Tool Comparison

Stage	Preferred	Alternatives	Best For	Notes
Local Contrast	VeraLux Revela	CLAHE, Native	Texture enhancement	Use conservatively
Color Grading	VeraLux Vectra	Curves, Saturation	Final polish	Protect star colors
HDR	Native HDR	Masked processing	High dynamic range	Only when beneficial
Export	16-bit TIFF	PNG, JPEG	Archive then publish	Never overwrite master

Common Finishing Mistakes

- Over-saturating colors.
- Applying excessive local contrast.
- Destroying star color during finishing.
- Exporting only compressed formats.

Workflow Decision Guide

Goal	Recommended Tool	Alternative
Natural color	SPCC + Vectra	PCC + Curves
Maximum texture	Revela	CLAHE
Publication master	16-bit TIFF	FITS archive

Notes

The last five percent of processing often takes the longest. Resist the temptation to keep adjusting an image simply because the tools are available. Know when to stop.

Key Takeaways

- Finish with restraint.
- Preserve natural color.
- Archive a full-quality master.
- Every adjustment should have a purpose.

Chapter 10 – Troubleshooting and Image Recovery

Introduction

Every astrophotographer encounters imperfect data. Successful image processing depends not on avoiding problems entirely, but on recognizing them early, identifying their causes, and choosing the least destructive corrective action.

Diagnose Before You Correct

Inspect the image systematically. Determine whether the problem originated during acquisition, calibration, stacking, or processing. Correct the cause rather than masking the symptom.

Common Problems

Residual gradients, clipped histograms, bloated stars, ringing, chromatic imbalance, walking noise, satellite trails, poor calibration, and over-processing are among the most common issues encountered.

Recovery Philosophy

Whenever possible, return to the earliest stage where the problem first appears. Avoid stacking multiple corrective operations on top of one another.

Troubleshooting Matrix

Symptom	Likely Cause	Recommended Tool	Alternative
Residual Gradient	Sky glow / poor flats	AutoBGE	GraXpert AI, VeraLux Nox
Soft Detail	Seeing / focus	RC-Astro BlurX	SyQon Parallax, Native Deconvolution
Excess Noise	Insufficient integration	RC-Astro NoiseX	SyQon Prism, Silentium
Star Halos	Over-processing	Reduce sharpening	Reprocess from earlier stage
Color Cast	Poor calibration	SPCC	PCC, Manual White Balance
Dark Halos	Aggressive star removal	Recombine carefully	Reduce star mask strength

Validation Workflow

- Review histogram after every major stage.
- Compare to the previous saved version.
- Inspect at Fit, 100%, and 200% zoom.
- Verify star color, background neutrality, and faint detail.

- Archive milestone versions before major changes.

Notes

If something suddenly looks dramatically better after a single processing step, stop and ask why. The best improvements are usually subtle and repeatable.

Core Processing Reference

The following reference summarizes the canonical workflow adopted throughout this companion.

Stage	Default	Alternatives	Validation	Related Chapters
Background Extraction	AutoBGE	GraXpert AI, VeraLux Nox	Neutral background	4, 11–21
Color Calibration	SPCC	PCC, White Balance	Natural stars	4
Stretch	GHS	Statistical, Asinh	Histogram preserved	5
Sharpen	RC-Astro BlurX	Parallax, Native	No halos	6
Denoise	RC-Astro StarX	Prism, Silentium	Dust retained	7
Stars	StarX	StarNet, Native	Natural recombination	8
Finishing	Revela + Vectra	Native tools	Subtle appearance	9

Workflow Profiles Summary

Profile	Purpose	Typical Use
Fast Preview	Rapid evaluation	Session review
Standard	Balanced workflow	Routine imaging
Premium	Enhanced quality	Portfolio images
Maximum Aesthetic	Highest visual quality	Showcase work
Scientific	Maximum fidelity	Measurement/documentation

Core Processing Checklist

- Good calibration before enhancement.
- Independent sharpening and denoising.
- Validate after each major stage.
- Save 16-bit archival master.
- Export separate web and print versions.

Chapter 11 – Galaxy Processing

Introduction

Galaxy processing emphasizes preserving fine dust lanes, realistic star colors, and a controlled dynamic range while revealing faint outer structures.

Scientific Background

Galaxies contain stellar populations, dust, gas, and active nuclei. Spiral arms and dust lanes often require different treatment than the bright core.

Acquisition Strategy

Prioritize long integration, accurate focus, and dark skies. Preserve bright cores by avoiding excessive exposure lengths.

Recommended Workflow

Calibrate → Background Extraction → SPCC → Linear Validation → Stretch → Sharpen → Denoise → Optional Star Management → Local Contrast → Color Grade → Export.

Galaxy Tool Selection

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, VeraLux Nox	Most datasets	Conservative extraction
Sharpen	RC-Astro BlurX	SyQon Parallax, Native RL	Dust lanes	Avoid halos
Denoise	RC-Astro CC Denoise	Prism, Silentium	Background	Protect faint galaxies
Contrast	VeraLux Revela	CLAHE	Dust texture	Use lightly
Finish	Vectra	Curves	Final polish	Preserve star color

Validation Checklist

- Core not clipped
- Dust lanes natural
- Outer halo preserved
- Stars retain color
- No sharpening artifacts

Notes

Galaxies reward patience. The temptation is to force contrast into faint spiral arms, but subtle processing usually produces the most believable result.

Process the same galaxy using GHS and Statistical Stretch. Compare dust lane contrast, core preservation, and outer halo visibility.

Chapter 12 – Emission Nebula Processing

Introduction

Emission nebulae often contain enormous dynamic range and intricate hydrogen and oxygen structures. The objective is to reveal filamentary detail while maintaining smooth backgrounds.

Scientific Background

Hydrogen-alpha, Oxygen III, and Sulfur II emissions originate from different physical processes. Understanding the source of the signal helps guide color mapping and contrast decisions.

Acquisition Strategy

Long integrations, dithering, and accurate calibration are essential. Broadband and narrowband datasets require different color strategies.

Recommended Workflow

Background Extraction → SPCC/PCC → Stretch → Sharpen → Denoise → Optional Starless Processing → Local Contrast → Color Grading → Star Recombination.

Emission Nebula Tool Selection

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, Nox	Gradient removal	Avoid removing faint nebula
Sharpen	BlurX	SyQon Parallax	Filaments	Moderate settings
Denoise	SyQon Prism	CC Denoise, Silentium	Faint emission	Preserve structure
Stars	StarX	StarNet, Native	Dense fields	Temporary removal
Finish	Revela + Vectra	Native tools	Texture & color	Subtle enhancement

Decision Guide

Use temporary starless processing when stars compete with filamentary structure. Retain stars throughout for smaller nebulae or scientific presentations.

Common Mistakes

- Oversaturating Hα
- Removing faint O III during denoise
- Overusing local contrast
- Leaving recombination halos

Notes

The most memorable nebula images are rarely the most saturated. They are the ones that preserve the delicate transitions between bright emission and the surrounding darkness.

Compare a conventional workflow with a temporary starless workflow and evaluate filament visibility, star color, and overall balance.

Chapter 13 – Reflection Nebula Processing

Introduction

Reflection nebulae are illuminated by starlight scattered from interstellar dust. Their beauty lies in delicate blue tones and extremely subtle dust structures that require restrained processing.

Scientific Background

Unlike emission nebulae, reflection nebulae shine by scattering light rather than emitting it. Their faint blue dust is easily overwhelmed by aggressive stretching or saturation.

Acquisition Strategy

Dark skies, long integration, excellent flats, and careful focus are essential. Reflection nebulae benefit more from total integration time than aggressive processing.

Recommended Workflow

Calibrate → AutoBGE → Background Neutralization → SPCC → Gentle GHS/Statistical Stretch → Light Sharpen → Conservative Denoise → Optional Star Separation → Local Contrast → Finish.

Tool Selection Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, VeraLux Nox	Minor gradients	Protect faint dust
Sharpen	RC-Astro BlurX	SyQon Parallax, Native RL	Fine dust	Use lightly
Denoise	SyQon Prism	RC-Astro NoiseX, Silentium	Low-contrast dust	Avoid plastic appearance
Stars	StarX	StarNet, Native	Optional	Only if dust benefits
Finish	Revela + Vectra	Native Curves	Final polish	Keep blue tones natural

Validation Checklist

- Blue reflection remains natural.
- Background not clipped.
- Faint dust preserved.
- No halos around bright stars.
- Color remains believable.

Notes

Reflection nebulae reward restraint. If the processing draws more attention than the dust itself, it is usually time to step back.

Compare GHS and Statistical Stretch on the same reflection nebula and evaluate dust visibility, star color, and background smoothness.

Chapter 14 – Dark Nebula Processing

Introduction

Dark nebulae are defined by the absence of light. Success comes from preserving subtle absorption features without crushing the surrounding background.

Scientific Background

Dark nebulae obscure the Milky Way behind them. Their structure is recognized by delicate tonal differences rather than bright emission.

Acquisition Strategy

Long integrations, excellent flat calibration, and gradient-free data are essential. Weak acquisition cannot be rescued by processing.

Recommended Workflow

Calibrate → Conservative Background Extraction → SPCC → Gentle Stretch → Minimal Sharpen → Conservative Denoise → Local Contrast → Finish.

Tool Selection Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, VeraLux Nox	Preserve absorption	Avoid overcorrection
Stretch	Statistical Stretch	Gentle GHS	Low contrast	Protect blacks
Sharpen	Light BlurX	Parallax, Native	Selective detail	Minimal use
Denoise	SyQon Prism	Silentium, CC Denoise	Dust preservation	Mask if needed
Finish	Vectra	Native Curves	Subtle grading	Natural appearance

Common Mistakes

- Crushing the black point.
- Removing genuine dust as a gradient.
- Oversharpening diffuse absorption.
- Heavy denoising that erases subtle structure.

Validation Checklist

- Absorption lanes remain visible.
- Background remains smooth.
- No clipped shadows.
- Natural Milky Way appearance.

Notes

Dark nebulae taught me that sometimes the most beautiful subject is what isn't illuminated. Preserving subtlety is the highest compliment you can pay the data.

Process the same dark nebula twice—once with aggressive contrast and once conservatively. Compare realism, dust continuity, and background quality.

Chapter 15 – Planetary Nebula Processing

Introduction

Planetary nebulae are compact objects with bright cores, intricate shells, and often extremely faint outer halos. The challenge is preserving dynamic range while revealing delicate internal structure.

Scientific Background

Planetary nebulae are formed when Sun-like stars expel their outer layers. Emission from ionized oxygen, hydrogen, and other elements creates rich color and complex internal detail.

Acquisition Strategy

Excellent focus, stable guiding, generous integration, and appropriate image scale are critical. Consider drizzle integration for undersampled datasets.

Recommended Workflow

Calibrate → Background Extraction → SPCC → Linear Validation → Gentle Stretch → Masked Sharpening → Conservative Denoise → Local Contrast → Color Refinement → Export.

Tool Selection Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, VeraLux Nox	Gradient removal	Preserve faint halo
Sharpen	RC-Astro BlurX	SyQon Parallax, Native RL	Fine shell detail	Mask bright core
Denoise	RC-Astro CC Denoise	SyQon Prism, Silentium	Background	Avoid smoothing filaments
Contrast	VeraLux Revela	CLAHE	Shell texture	Use sparingly
Finish	Vectra	Curves	Color refinement	Maintain OIII balance

Validation Checklist

- Core not clipped
- Outer halo retained
- Internal shells resolved
- Natural star colors
- No ringing artifacts

Notes

Planetary nebulae remind me that even small objects can contain astonishing complexity. The goal is not to make them look larger, but to reveal the remarkable detail already present.

Compare AI sharpening with Richardson–Lucy deconvolution on the same planetary nebula and evaluate shell detail, star profiles, and halo preservation.

Chapter 16 – Globular Cluster Processing

Introduction

Globular clusters are dense stellar systems whose beauty depends on resolving individual stars while preserving the natural brightness gradient from core to halo.

Scientific Background

Globular clusters contain some of the oldest stars in the Milky Way. Their crowded cores demand careful processing to avoid merging stellar profiles.

Acquisition Strategy

Use precise focus, avoid saturating the core, and collect sufficient integration to maintain color in faint outer members.

Recommended Workflow

Calibrate → Background Extraction → SPCC → Stretch → Moderate Sharpen → Light Denoise → Curves → Final Color Refinement.

Tool Selection Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI	Background cleanup	Minimal correction
Sharpen	BlurX	Native RL, Parallax	Resolve stars	Avoid halos
Denoise	Light CC Denoise	Prism, Native	Background only	Protect stellar profiles
Stars	Retain	Minimal reduction	Clusters	Stars are the subject
Finish	Vectra	Native Curves	Natural color	Preserve stellar diversity

Common Mistakes

- Over-sharpening until halos appear.
- Reducing star sizes excessively.
- Clipping the bright core.
- Over-saturating stellar colors.

Validation Checklist

- Core resolved
- Natural stellar colors
- Outer halo preserved

- No artificial star shapes

Notes

A globular cluster should resemble a three-dimensional sphere suspended in space. If every star looks identical, the processing has gone too far.

Produce two versions: one emphasizing maximum sharpness and one emphasizing natural stellar appearance. Compare which better represents the original data.

Chapter 17 – Integrated Flux Nebula (IFN) Processing

Introduction

Integrated Flux Nebula (IFN) is among the faintest astrophotography targets. Success depends on preserving extremely low-contrast dust while resisting the temptation to overprocess.

Scientific Background

IFN is illuminated by the integrated light of the Milky Way rather than nearby stars. The structures are broad, faint, and only slightly brighter than the surrounding sky.

Acquisition Strategy

Dark skies, excellent transparency, long integration times, meticulous flat calibration, and careful dithering are essential. Acquisition quality is more important than aggressive processing.

Recommended Workflow

Calibrate → Conservative AutoBGE → SPCC → Statistical Stretch or Gentle GHS → Minimal Sharpen → Conservative Denoise → Subtle Local Contrast → Final Color Refinement.

Tool Selection Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, VeraLux Nox	Preserve faint dust	Avoid overcorrection
Stretch	Statistical Stretch	Gentle GHS	Ultra-low contrast	Maintain histogram
Sharpen	None or Very Light BlurX	SyQon Parallax	Selective enhancement	Less is more
Denoise	SyQon Prism	CC Denoise, Silentium	Protect faint IFN	Mask when possible
Finish	Vectra	Native Curves	Subtle color	Natural appearance

Validation Checklist

- Background not clipped
- Faint IFN remains continuous
- No artificial texture
- Natural star colors
- Histogram preserved

Notes

IFN taught me that sometimes the finest images whisper instead of shout. Preserve the faintest dust and let the viewer discover it.

Compare an aggressively processed IFN image with a restrained version. Evaluate dust continuity, background smoothness, and realism.

Chapter 18 – Comet Processing

Introduction

Comets are unique because the comet moves while the stars remain fixed. Effective processing often requires separate comet-aligned and star-aligned integrations.

Scientific Background

A comet consists of a nucleus, coma, dust tail, and ion tail. The apparent motion across the sky introduces challenges not encountered with deep-sky objects.

Acquisition Strategy

Use exposure lengths appropriate to the comet's apparent motion. For fast-moving comets, acquire sufficient frames for both comet-aligned and star-aligned stacks.

Recommended Workflow

Calibrate → Build Star Stack → Build Comet Stack → Background Extraction → Stretch → Moderate Sharpen → Light Denoise → Blend → Final Color Refinement.

Tool Selection Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI	Gradient removal	Protect tails
Sharpen	BlurX	Parallax, Native	Coma detail	Moderate settings
Denoise	CC Denoise	Prism, Silentium	Background	Do not soften tails
Blend	Masked Blend	PixelMath, Native	Dual-stack workflow	Align carefully
Finish	Vectra	Native Curves	Natural color	Preserve ion tail

Common Mistakes

- Misaligning comet and star stacks.
- Oversharpening the coma.
- Clipping faint dust or ion tails.
- Over-saturating the nucleus.

Validation Checklist

- Sharp nucleus
- Natural stars
- Dust tail continuous
- Ion tail preserved

- No blend artifacts

Notes

Every comet is a visitor with its own personality. The goal is not to force it into a standard deep-sky workflow, but to tell the story of its journey.

Process one version from a star-aligned stack only and another using separate comet and star stacks. Compare realism and tail definition.

Chapter 19 – Broadband OSC Processing

Introduction

One-shot color (OSC) cameras have transformed astrophotography by simplifying acquisition while delivering excellent scientific and aesthetic results. Success depends on maximizing signal-to-noise ratio and preserving natural color throughout processing.

Scientific Background

Broadband OSC cameras record red, green, and blue information simultaneously through a Bayer matrix. Debayering reconstructs full-color images, making careful calibration essential.

Acquisition Strategy

Use accurate focus, dithering, quality flats, and sufficient total integration. Broadband imaging benefits greatly from dark skies and good transparency.

Recommended Workflow

Calibrate → Debayer → Register → Integrate → AutoBGE → SPCC → Gentle Stretch → Sharpen → Denoise → Optional Star Management → Local Contrast → Color Refinement → Export.

Broadband OSC Tool Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, VeraLux Nox	General gradients	Default choice
Stretch	GHS	Statistical Stretch, Asinh	Mixed targets	Protect highlights
Sharpen	RC-Astro BlurX	SyQon Parallax, Native	General enhancement	Moderate settings
Denoise	RC-Astro CC Denoise	Prism, Silentium	Background	Preserve faint signal
Finish	Revela + Vectra	Native tools	Final polish	Maintain natural color

Validation Checklist

- Color calibration accurate
- Histogram preserved
- Natural star colors
- Minimal gradients
- No oversaturation

Notes

After decades of monochrome imaging and filter wheels, modern OSC cameras still amaze me. The simplicity of acquisition lets you spend more time under the stars and less time changing filters.

Process the same OSC dataset using GHS and Statistical Stretch. Compare color fidelity, faint nebulosity, and star color.

Chapter 20 – Dual-Band and Narrowband OSC Processing

Introduction

Dual-band and narrowband filters allow OSC cameras to image emission nebulae effectively even under light-polluted skies. Careful processing is required to preserve both hydrogen and oxygen structures.

Scientific Background

Dual-band filters isolate Hα and O III emission while rejecting much of the surrounding light pollution. Narrowband processing emphasizes structure over broadband color realism.

Acquisition Strategy

Use long integrations, accurate calibration, and consistent focus. Narrowband datasets often benefit from additional total exposure compared with broadband imaging.

Recommended Workflow

Calibrate → Integrate → AutoBGE → Color Calibration (if appropriate) → Stretch → Filament Enhancement → Denoise → Optional Starless Processing → Recombine → Final Color Refinement.

Dual-Band Tool Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, VeraLux Nox	Gradient removal	Protect faint emission
Sharpen	RC-Astro BlurX	SyQon Parallax	Filaments	Moderate application
Denoise	SyQon Prism	CC Denoise, Silentium	Weak emission	Retain structure
Stars	StarX	StarNet, Native	Dense fields	Temporary removal
Finish	Revela + Vectra	Native tools	Texture & color	Avoid oversaturation

Common Mistakes

- Over-emphasizing Hα at the expense of O III.
- Removing faint filaments with excessive denoising.
- Leaving halos after star recombination.
- Oversaturating narrowband colors.

Validation Checklist

- Filaments remain continuous

- Balanced Hα/O III presentation
- Natural-looking stars
- Smooth background
- No recombination artifacts

Notes

Modern dual-band filters have opened opportunities that simply did not exist when I began. They allow remarkable work from locations that would once have been unsuitable for deep-sky imaging.

Compare a standard workflow with a temporary starless workflow on the same dual-band dataset and evaluate filament visibility and overall balance.

Chapter 21 – Mono LRGB Processing

Introduction

Monochrome LRGB imaging remains the benchmark for extracting maximum spatial resolution, precise color, and faint detail. Separating luminance from color provides tremendous flexibility during processing.

Scientific Background

A monochrome sensor records every incoming photon without the Bayer interpolation required by OSC cameras. Separate L, R, G, and B filters allow each channel to be optimized independently.

Acquisition Strategy

Prioritize luminance integration, maintain critical focus for every filter, refocus as temperature changes, and collect complete calibration data for each channel.

Recommended Workflow

Calibrate → Integrate L/R/G/B → AutoBGE → SPCC/PCC → Process Luminance → Process RGB → LRGB Combination → Local Contrast → Color Refinement → Export.

LRGB Tool Matrix

Stage	Recommended	Alternatives	Best Use	Notes
Background	AutoBGE	GraXpert AI, VeraLux Nox	All channels	Treat channels independently
Sharpen	RC-Astro BlurX	SyQon Parallax, Richardson–Lucy	Luminance	Avoid RGB oversharpening
Denoise	RC-Astro CC Denoise	SyQon Prism, Silentium	RGB & background	Preserve color
LRGB Combine	Native LRGB	PixelMath	Final merge	Inspect color balance
Finish	Revela + Vectra	Native Curves	Final polish	Natural appearance

Validation Checklist

- Luminance crisp without halos
- RGB smooth and natural
- Balanced color after combination
- No clipped highlights
- Faint structures retained

Notes

When I first built a Cookbook CCD camera, collecting separate filters felt revolutionary. Today's software makes LRGB processing far easier, but the underlying principles remain unchanged: collect the best luminance you can and let the color support—not compete with—the detail.

Create one image using only RGB and a second using a full LRGB combination. Compare resolution, noise, and color fidelity.

Chapter 22 – Complete End-to-End Processing Case Study

Purpose

This capstone chapter demonstrates how all of the concepts presented throughout the companion work together during a complete imaging project.

Project Planning

Select target, review moon phase, weather, altitude, and imaging goals.

Acquisition

Capture calibrated light frames with dithering, autofocus, and sufficient integration.

Calibration & Integration

Produce a clean master image using consistent calibration.

Linear Processing

Crop, background extraction, color calibration, and validation.

Nonlinear Processing

Stretch carefully while preserving dynamic range.

Enhancement

Sharpen, denoise, manage stars, and enhance local contrast.

Finishing

Color grading, final inspection, archive 16-bit master, export web and print versions.

Complete Workflow Summary

Stage	Primary Tool	Alternatives	Validation
Background	AutoBGE	GraXpert AI, Nox	Neutral background
Stretch	GHS	Statistical, Asinh	Histogram preserved
Sharpen	BlurX	Parallax, RL	No halos
Denoise	CC Denoise	Prism, Silentium	Dust retained
Stars	StarX	StarNet	Natural recombination
Finish	Revela + Vectra	Native tools	Natural final image

Final Publication Checklist

- Compare with original master.
- Inspect at Fit, 100%, and 200%.
- Verify background neutrality.
- Confirm natural star colors.
- Archive the master image.
- Document major processing decisions.

Notes

Across six decades, the equipment has changed dramatically, but the satisfaction of revealing light that has traveled across the universe has not. Technology evolves; curiosity endures.

Closing Thoughts

The purpose of this companion is not to prescribe a single workflow but to help you understand your data, evaluate your options, and choose the right tools for each image. Master the principles, remain curious, and continue refining both your technique and your eye.

Workflow Library

Workflow Standard v1.0
Foundation Specification for the Second Edition Companion Library

Purpose

This standard defines the conventions, terminology, processing philosophy, and quality requirements for every workflow in the Chuck Faranda Professional Workflow Library. All Seti Astro Suite Pro and Siril Workflow Assistant files will conform to this specification.

Workflow Philosophy

• Preserve astronomical signal before enhancing aesthetics.
• Use AI where it provides measurable benefits, not by default.
• Favor scientifically accurate color.
• Inspect results after every major processing stage.
• Always provide practical alternatives where appropriate.
• Avoid destructive processing whenever possible.

Preferred Processing Order

1. Crop / Geometry
2. Calibration Verification
3. Background Extraction
4. Color Calibration
5. Linear Validation
6. Stretch
7. Sharpening
8. Noise Reduction
9. Star Management (optional)
10. Local Contrast
11. Color Enhancement
12. Final Validation
13. Archive

Preferred Tool Hierarchy

Background: AutoBGE → GraXpert AI → VeraLux Nox → Manual
Color Calibration: SPCC → PCC → Manual White Balance
Stretch: Generalized Hyperbolic Stretch → Statistical Stretch → Asinh → Histogram
Sharpening: BlurXTerminator (RC-Astro) → SyQon Parallax → Native Deconvolution

Noise Reduction: NoiseXTerminator (RC-Astro) → SyQon Prism → VeraLux Silentium → Native
Star Management: StarXTerminator (RC-Astro) → SyQon Starless → StarNet
Local Contrast: VeraLux Revela → CLAHE → Wavelets
Color Enhancement: VeraLux Vectra → Curves → Saturation

Required Metadata

Every workflow shall include:
• Workflow title
• Author: Chuck Faranda
• Library name
• Edition
• Workflow version
• Compatible software
• Revision history

Required Step Documentation

Each processing step should identify:
• Purpose
• Preferred tool
• Alternatives
• Recommended use
• When not to use
• Common mistakes
• Validation checkpoint
• Chapter reference

Quality Standards

All workflows should:
• Preserve faint dust and IFN where present.
• Avoid clipped highlights and shadows.
• Preserve natural star colors.
• Minimize halos and artifacts.
• Save a 16-bit TIFF master.

Confidence Ratings

★★★★★ Strongly recommended
★★★★☆ Recommended
★★★☆☆ Situational

★★☆☆☆ Specialized
★☆☆☆☆ Rarely recommended

Naming Convention

Chuck Faranda Professional Workflow Library
Examples:
• Chuck Faranda Professional Master Workflow
• Chuck Faranda Professional Galaxy Workflow
• Chuck Faranda Professional IFN Workflow

Publication Goal

The workflow library is intended to provide import-ready workflows for Seti Astro Suite Pro and Siril Workflow Assistant while maintaining a consistent processing philosophy across the entire companion library.

Workflow Standard v1.1 – Addendum

Workflow Scope

These workflows are intended for deep-sky astrophotography processed with Seti Astro Suite Pro and/or the Siril Workflow Assistant. They are optimized for calibrated, registered, and stacked data from one-shot color (OSC) and monochrome cameras. Individual workflows may recommend modifications based on target type, imaging conditions, filter selection, image scale, and overall data quality.

Expanded Workflow Philosophy

• Preserve scientific integrity whenever practical.

• Preserve astronomical signal before enhancing aesthetics.

• Prefer incremental adjustments over aggressive single-step processing.

• Use AI where it provides measurable benefits.

• Validate results after every major processing stage.

• Maintain a repeatable, documented workflow.

Background Extraction Guidance

AutoBGE – Preferred default for most images.

GraXpert AI – Recommended when complex gradients remain.

VeraLux Nox – Useful for difficult background correction.

Manual Extraction – Reserved for advanced users and specialized situations.

AI Tool Philosophy

RC-Astro, SyQon, VeraLux, GraXpert, and other AI-assisted tools are intended to enhance—not replace—careful image processing. Apply AI tools conservatively, inspect results after every major operation, and preserve the integrity of the original astronomical data.

Workflow Validation Checklist

☐ Histogram verified

☐ Background neutral

☐ No clipped highlights

☐ No clipped shadows

☐ Natural star colors preserved

☐ No sharpening halos

☐ Faint dust / IFN preserved where present

☐ Image inspected at Fit

☐ Image inspected at 100%

☐ 16-bit TIFF master archived

Confidence Rating Definitions

★★★★★ Default recommendation for most datasets.

★★★★☆ Recommended for many datasets.

★★★☆☆ Situational; use when conditions warrant.

★★☆☆☆ Specialized workflow.

★☆☆☆☆ Experimental or rarely required.

Version Compatibility

These workflows are designed for current releases of Seti Astro Suite Pro and the Siril Workflow Assistant at the time of publication. Future software updates may require minor revisions while preserving the overall processing philosophy.

Workflow

Workflow Reference Manual

This reference manual provides a concise, one-page summary of each workflow in the Chuck Faranda Professional Workflow Library. It is intended as a quick-reference companion to The Complete Siril & Seti Astro Suite Pro Processing Companion. Download the workflow Json files here.

CFPW-001 — Master Workflow

Primary Target	General purpose
Background Extraction	AutoBGE
Color Calibration	SPCC
Preferred Stretch	GHS
BlurXTerminator	BlurXTerminator
NoiseXTerminator	NoiseXTerminator
StarXTerminator	Optional
Primary Goal	All-purpose reference workflow.
Validation	Inspect histogram, background neutrality, star colors, halos, and archive a 16-bit TIFF master.
Book Cross-Reference	See the corresponding target chapter and workflow discussion.

Recommended AI Tools:

BlurXTerminator, NoiseXTerminator, and StarXTerminator (RC-Astro) as appropriate; SyQon Parallax/Prism and VeraLux tools where recommended by the workflow.

CFPW-101 — Galaxy

Primary Target	Broadband galaxies
Background Extraction	AutoBGE
Color Calibration	SPCC
Preferred Stretch	GHS
BlurXTerminator	Moderate
NoiseXTerminator	Light
StarXTerminator	Optional
Primary Goal	Preserve dust lanes and galaxy halos.
Validation	Inspect histogram, background neutrality, star colors, halos, and archive a 16-bit TIFF master.
Book Cross-Reference	See the corresponding target chapter and workflow discussion.

Recommended AI Tools:

BlurXTerminator, NoiseXTerminator, and StarXTerminator (RC-Astro) as appropriate; SyQon Parallax/Prism and VeraLux tools where recommended by the workflow.

CFPW-102 — Emission Nebula

Primary Target	Hα/OIII/SII
Background Extraction	AutoBGE
Color Calibration	SPCC
Preferred Stretch	GHS
BlurXTerminator	15–25%
NoiseXTerminator	Light
StarXTerminator	Recommended
Primary Goal	Enhance filaments while preserving faint emission.
Validation	Inspect histogram, background neutrality, star colors, halos, and archive a 16-bit TIFF master.
Book Cross-Reference	See the corresponding target chapter and workflow discussion.

Recommended AI Tools:

BlurXTerminator, NoiseXTerminator, and StarXTerminator (RC-Astro) as appropriate; SyQon Parallax/Prism and VeraLux tools where recommended by the workflow.

CFPW-103 — Reflection Nebula

Primary Target	Broadband reflection
Background Extraction	AutoBGE
Color Calibration	SPCC
Preferred Stretch	GHS
BlurXTerminator	10–20%
NoiseXTerminator	Very Light
StarXTerminator	Usually No
Primary Goal	Protect blue dust and IFN.
Validation	Inspect histogram, background neutrality, star colors, halos, and archive a 16-bit TIFF master.
Book Cross-Reference	See the corresponding target chapter and workflow discussion.

Recommended AI Tools:

BlurXTerminator, NoiseXTerminator, and StarXTerminator (RC-Astro) as appropriate; SyQon Parallax/Prism and VeraLux tools where recommended by the workflow.

CFPW-104 — Dark Nebula / IFN

Primary Target	Broadband dust
Background Extraction	AutoBGE
Color Calibration	SPCC
Preferred Stretch	Statistical Stretch
BlurXTerminator	0–10%
NoiseXTerminator	Minimal
StarXTerminator	No
Primary Goal	Maximum preservation of faint dust.
Validation	Inspect histogram, background neutrality, star colors, halos, and archive a 16-bit TIFF master.
Book Cross-Reference	See the corresponding target chapter and workflow discussion.

Recommended AI Tools:

BlurXTerminator, NoiseXTerminator, and StarXTerminator (RC-Astro) as appropriate; SyQon Parallax/Prism and VeraLux tools where recommended by the workflow.

CFPW-105 — Planetary Nebula

Primary Target	Compact nebulae
Background Extraction	AutoBGE
Color Calibration	SPCC
Preferred Stretch	GHS
BlurXTerminator	25–40%
NoiseXTerminator	Light
StarXTerminator	Optional
Primary Goal	Reveal shells without clipping cores.
Validation	Inspect histogram, background neutrality, star colors, halos, and archive a 16-bit TIFF master.
Book Cross-Reference	See the corresponding target chapter and workflow discussion.

Recommended AI Tools:

BlurXTerminator, NoiseXTerminator, and StarXTerminator (RC-Astro) as appropriate; SyQon Parallax/Prism and VeraLux tools where recommended by the workflow.

CFPW-106 — Globular Cluster

Primary Target	Dense star clusters
Background Extraction	AutoBGE
Color Calibration	SPCC
Preferred Stretch	GHS
BlurXTerminator	15–25%
NoiseXTerminator	Very Light
StarXTerminator	No
Primary Goal	Preserve stellar colors and resolve cores.
Validation	Inspect histogram, background neutrality, star colors, halos, and archive a 16-bit TIFF master.
Book Cross-Reference	See the corresponding target chapter and workflow discussion.

Recommended AI Tools:

BlurXTerminator, NoiseXTerminator, and StarXTerminator (RC-Astro) as appropriate; SyQon Parallax/Prism and VeraLux tools where recommended by the workflow.

CFPW-107 — Comet

Primary Target	Comets
Background Extraction	AutoBGE
Color Calibration	SPCC
Preferred Stretch	GHS
BlurXTerminator	15–25%
NoiseXTerminator	Light
StarXTerminator	Optional
Primary Goal	Protect coma and faint ion/dust tails.
Validation	Inspect histogram, background neutrality, star colors, halos, and archive a 16-bit TIFF master.
Book Cross-Reference	See the corresponding target chapter and workflow discussion.

Recommended AI Tools:

BlurXTerminator, NoiseXTerminator, and StarXTerminator (RC-Astro) as appropriate; SyQon Parallax/Prism and VeraLux tools where recommended by the workflow.

Workflow Guide

Workflow User Guide
Second Edition

This guide accompanies The Complete Siril & Seti Astro Suite Pro Processing Companion and explains how to install, import, customize, and use the Chuck Faranda Professional Workflow Library.

1. Introduction

The workflow library provides import-ready workflows for Seti Astro Suite Pro and the Siril Workflow Assistant. Each workflow is optimized for a specific target type or processing objective while following a consistent philosophy of preserving astronomical signal and applying AI tools judiciously.

2. Workflow Categories

• CFPW-001: Master Workflows
• CFPW-101–107: Target Workflows (Galaxy, Emission, Reflection, Dark Nebula/IFN, Planetary, Globular, Comet)
• CFPW-201 Series: Acquisition Workflows
• CFPW-301 Series: Quality Profiles

3. Installing Workflows

Copy the supplied JSON files into the appropriate workflow folder for your software, or use each application's Import Workflow feature if available. Keep the original files as a backup before editing.

4. Importing into Seti Astro Suite Pro

Open the Workflow Assistant, choose Import Workflow, select the desired JSON file, and verify that the imported workflow appears with the expected title and processing steps.

5. Importing into Siril Workflow Assistant

Open the Workflow Assistant, import the JSON file, and confirm that the default selections and notes are loaded correctly.

6. Choosing the Correct Workflow

Select the workflow that best matches your imaging target. For example, use the Galaxy workflow for broadband galaxies, the Emission Nebula workflow for Hα/OIII/SII targets, and the Dark Nebula workflow when preserving IFN and faint dust is the primary goal.

7. AI Tool Integration

Preferred tools:
• BlurXTerminator (RC-Astro)
• NoiseXTerminator (RC-Astro)
• StarXTerminator (RC-Astro)

Alternatives include SyQon Parallax, SyQon Prism, GraXpert AI, VeraLux Nox, VeraLux Revela, and VeraLux Vectra where appropriate.

8. Customizing Workflows

Modify copies of workflows rather than the originals. Record changes, keep version numbers, and validate results after each major processing step.

9. Validation Checklist

Before exporting, verify:
• Histogram not clipped
• Neutral background
• Natural star colors
• No sharpening halos
• Faint dust preserved
• 16-bit TIFF master archived

10. Troubleshooting

If results differ from expectations, verify calibration quality, background extraction, color calibration, AI tool strength, and stretch settings before making further adjustments.

11. Best Practices

Work incrementally, save intermediate versions, compare against the original image frequently, and avoid aggressive processing that removes real astronomical signal.

12. Frequently Asked Questions

Q: Should I always use AI tools?
A: No. Use them where they provide measurable improvement while preserving data.

Q: Which workflow should I start with?
A: The Master Workflow or the workflow matching your target type.

Q: Can I edit these workflows?
A: Yes. Create your own versions while preserving the originals.

Appendix A – RC-Astro Tool Reference

RC-Astro tools are widely regarded as industry-leading AI-assisted utilities for sharpening, denoising, and star management. This appendix summarizes when and how to use them.

BlurX Sharpen

Purpose: Recover fine detail while preserving natural appearance.

Best Targets

- Galaxies
- Planetary nebulae
- Broadband OSC
- Mono LRGB

Strengths and Limitations

Strengths: Excellent AI detail recovery, easy to use.
Limitations: Commercial software; avoid aggressive settings on noisy data.

Workflow Integration

Typical Workflow Position: After stretching and before final denoising.

NoiseX Denoise

Purpose: Reduce random noise while preserving faint structure.

Best Targets

- Broadband
- Narrowband
- Reflection Nebulae
- IFN

Strengths and Limitations

Strengths: Excellent signal preservation.
Limitations: Excessive settings may soften microcontrast.

Workflow Integration

Typical Workflow Position: After sharpening, before finishing.

StarX

Purpose: Temporary or permanent star removal for independent processing.

Best Targets

- Emission Nebulae
- Dense Milky Way Fields

Strengths and Limitations

Strengths: High-quality star separation.
Limitations: Requires careful recombination to avoid halos.

Workflow Integration

Typical Workflow Position: After nonlinear processing.

RC-Astro Decision Matrix

Need	Recommended Tool	Alternative	Comments
Recover detail	BlurX	SyQon Parallax	General recommendation
Reduce noise	NoiseX	Prism, Silentium	Protect faint signal
Star removal	StarX	StarNet	Recombine carefully

Appendix B – SyQon Tool Reference

SyQon tools provide powerful AI-assisted alternatives with an emphasis on preserving subtle astronomical structure.

Parallax

Purpose: AI-assisted sharpening emphasizing fine filamentary and dust detail.

Best Targets

- Emission Nebulae
- Reflection Nebulae
- Galaxies

Strengths and Limitations

Strengths: Excellent texture preservation.
Limitations: Requires careful comparison against the original.

Workflow Integration

Workflow: Primary sharpening alternative to RC-Astro.

Prism

Purpose: Intelligent denoising designed to retain faint structures.

Best Targets

- IFN
- Reflection Nebulae
- Broadband
- Narrowband

Strengths and Limitations

Strengths: Outstanding preservation of weak signal.
Limitations: Use conservatively on already clean data.

Workflow Integration

Workflow: Denoising after stretch and sharpening.

Starless

Purpose: Star separation for independent processing.

Best Targets

- Nebulae

- Wide-field Milky Way

Strengths and Limitations

Strengths: Flexible starless workflows.
Limitations: Inspect recombination for artifacts.

Workflow Integration

Workflow: Optional stage before local contrast.

SyQon Decision Matrix

Need	Recommended Tool	Alternative	Comments
Fine filament detail	Parallax	BlurX	Excellent on nebulae
Preserve faint signal	Prism	CC Denoise	Ideal for IFN
Star separation	Starless	StarX	Evaluate recombination

Appendix Summary

Neither RC-Astro nor SyQon is universally superior. The Workflow Companion encourages selecting the tool that best matches the target, data quality, and desired outcome. Experience and objective comparison should guide every processing decision.

References

The following publications, organizations, and technical resources informed the concepts and best practices presented throughout this companion.

- The Deep-Sky Imaging Primer (3rd Edition) — Charles Bracken
- Lessons from the Masters — Ron Wodaski
- Handbook of Astronomical Image Processing — Richard Berry & James Burnell
- Star Ware — Philip Harrington
- CCD Astronomy – Christian Buil
- Outer Space Photography – Dr. Paul
- A manual of Advanced Celestial Photography – Brad Wallis & Robert Provin

Scientific Organizations

- NASA
- European Space Agency (ESA)
- International Astronomical Union (IAU)
- American Astronomical Society (AAS)

Software and Resources

Software	Primary Purpose	Notes
Siril	Calibration and image processing	Primary workflow platform
Seti Astro Suite Pro	AI-assisted image processing	Companion workflow platform
PixInsight	Advanced astrophotography processing	Professional processing environment
RC-Astro Tools	AI sharpening, denoising, star management	Commercial plug-ins
SyQon Tools	AI sharpening and denoising	Alternative enhancement tools
VeraLux Tools	Contrast, color, and finishing	Image refinement
GraXpert	Background extraction	Gradient removal
ASCOM Platform	Device interoperability	Observatory automation
N.I.N.A.	Image acquisition	Automation
ASTAP	Plate solving	Image analysis
PHD2	Guiding	Mount guiding

Acknowledgments

I gratefully acknowledges the developers, researchers, educators, and amateur astronomers whose software, publications, and willingness to share knowledge have advanced the field of astrophotography.

About Me

I have been actively engaged in amateur astronomy and astrophotography for more than six decades. My journey began in the mid-1960s with a 3-inch Unitron equatorial refractor and an Astrocam 220 plate camera, progressed through 35mm film astrophotography using Kodak Technical Pan film, and continued into the early CCD era with a self-built Cookbook CCD camera. Having witnessed the remarkable evolution from sheet film to today's AI-assisted image processing makes me marvel at today's modern astrophotography.

I am a retired firefighter and later retired from a second career as a City Manager for a municipality in South Florida. Throughout my professional life, astronomy remained a lifelong passion. Today, I live on Florida's Treasure Coast and own and operate 'Starry Nights Ranch' a private 40-acre astronomy complex near the Kissimmee Prairie Preserve (one of the darkest observing locations in Florida). The 'Ranch' serves as both a personal retreat and a development laboratory for my observatory automation projects for CCDASTRO, Inc. that provides design and observatory automation consulting services. CCDASTRO developed the m1OASYS and RRCI observatory automation systems, as well as ASCOM drivers for Takahashi Temma telescope mounts and other observatory equipment. The Company's goal has always been to simplify complex observatory operations while making advanced automation accessible to amateur astronomers.

I was married to my beloved wife, Donna, for 51 years. Together we raised two sons, Jason and Jonathan, and today I am the proud grandfather of five grandchildren: Trenten, Matthew, Benjamin, Ryan, and Amelia. Through this book I hope to help fellow astrophotographers make the most of today's extraordinary software tools while remembering that the ultimate objective is not simply creating beautiful images, but to also reveal the wonders of God's universe with accuracy.

A Note About Artificial Intelligence

AI is a powerful and very useful tool in our astrophotography. In fact, this document was developed through a collaborative process with OpenAI's ChatGPT. I created a framework via a Word document and AI supplied a manuscript version back after serving as proofreader and editor in an iterative fashion.

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